

Hospital Appointment Voice Agent

End-to-end n8n backend for a Vapi phone-call voice agent — technical & process documentation

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This report documents the n8n backend that powers a Vapi phone voice agent for a hospital front desk. Callers can check a doctor's availability, book a new appointment, cancel or reschedule an existing one, or ask for a human callback — entirely by voice — while n8n validates every request, keeps a Google Sheets record, and keeps hospital staff notified by email.

1. Executive Summary

Hospital front desks lose time to repetitive phone calls: “Is Dr. Ahmed free Thursday?”, “I need to cancel my 3pm”, “can someone call me back”. This backend gives a Vapi voice agent four callable tools that handle exactly these requests, each backed by validation logic so the agent can never book a conflicting slot, cancel someone else’s appointment, or skip caller confirmation before making a change.

Metric	Detail
Trigger	4 webhooks called directly by Vapi during a live phone call
Voice platform	Vapi — the caller talks to an AI agent, which calls these tools mid-conversation
Data store	Google Sheets — an Appointments tab and a Callbacks tab
Staff notifications	Gmail — sent on every new booking, cancellation, reschedule and callback request
Confirmation rule	Bookings and changes are rejected unless the caller explicitly confirmed them
Identity check	Cancel/reschedule requests must match an existing appointment ID AND phone number

2. Business Value

- Front desk staff freed up — routine scheduling calls are handled end-to-end by the voice agent, day or night.
- No double-bookings — every booking is checked against the doctor's existing confirmed slots before it's accepted.
- No unauthorized changes — an appointment can only be cancelled or rescheduled by someone who knows both the appointment ID and the phone number on file.
- Staff always in the loop — every booking, cancellation, reschedule and callback request lands in their inbox immediately.
- Safe fallback — when a request can't be resolved by voice, the agent logs a callback request instead of leaving the caller stuck.

3. Architecture Overview

Vapi's voice agent is configured with four tools, each pointing at its own n8n webhook. During a call, the agent decides which tool to call based on what the caller is asking for.

Tool / Webhook	Purpose
check_appointment_availability	Checks whether a doctor already has a confirmed slot at the requested date and time
book_appointment	Saves a caller-confirmed booking to Google Sheets and emails staff
manage_appointment	Finds an existing appointment by ID + phone, then cancels or reschedules it
request_human_callback	Logs a callback request and emails staff — not a live call transfer

4. Tool — check_appointment_availability

- 1 Receive the request**
A webhook accepts the doctor name, department, date and time the caller asked about.
- 2 Read existing appointments**
All rows are pulled from the Appointments sheet so they can be checked against the request.
- 3 Normalise & compare**
Dates and times are parsed into a consistent format regardless of how they arrive (e.g. “3pm” or “15:00”, “07/09/2026” or “2026-09-07”), then checked for a confirmed appointment with the same doctor, department, date and time that hasn't been cancelled.
- 4 Reply with a clear yes/no**
The response tells Vapi whether the slot is available, in a sentence the voice agent can speak directly to the caller — either confirming the slot or asking them to choose another time.

5. Tool — book_appointment

- 1 Validate the request**
Patient name, phone number, department, doctor, date and time must all be present, and the booking is rejected outright unless the caller explicitly confirmed it — the agent can't book on a guess.
- 2 Normalise the appointment**
A unique appointment ID is generated, and the incoming fields are standardised into the appointment record (defaulting appointment type to “new patient” and reason for visit to “Not provided” when the caller didn't give one).
- 3 Save & notify**
The appointment is appended to the Google Sheets Appointments tab as “confirmed”, and an email goes out to hospital staff with the full booking details.
- 4 Confirm back to the caller**
Vapi receives a confirmation message naming the patient, doctor, department, date and time, which the agent reads back to the caller along with their appointment ID.

6. Tool — manage_appointment

1 Validate the request

The action must be either “cancel” or “reschedule”, an appointment ID and phone number are required, the caller must have explicitly confirmed the change, and a reschedule additionally requires a new date and time.

2 Find the matching appointment

The Appointments sheet is searched for an active (non-cancelled) appointment whose ID and phone number both match what the caller provided — this double check prevents someone from cancelling an appointment that isn't theirs.

3 Not found? Stop safely

If no active appointment matches both the ID and the phone number, the agent tells the caller so instead of guessing which booking they meant.

4 Cancel

For a cancellation, the matching row's status is set to “cancelled”, a confirmation message is built, and staff are emailed that the appointment was cancelled.

5 Reschedule

For a reschedule, only the fields the caller actually changed (new date, time, doctor or department) are merged over the existing record, the row is updated in place, and staff are emailed the new details.

6 Confirm back to the caller

Vapi receives a status of cancelled, rescheduled or not_found along with a spoken-friendly message and the appointment ID.

7. Tool — request_human_callback

1 Capture the request

Patient name, phone number and the reason they need a callback are recorded, along with the conversation/call ID for traceability.

2 Log & notify

The request is appended to a separate Callbacks tab in Google Sheets as “pending”, and staff are emailed the patient's name, number and reason so a human can call them back.

3 Confirm back to the caller

Vapi tells the caller their callback request has been sent to hospital staff — this tool never transfers the live call, it only queues a follow-up.

8. Node-by-Node Map

Tool	Key nodes
check_appointment_availability	Appointment Availability Webhook → Read Appointments → Check Doctor Slot → Appointment Slot Available → Message - Available / Not Available → Respond to Vapi - Availability
book_appointment	Book Appointment Webhook → Validate Booking Request → Normalize Appointment Data → Log Appointment to Sheet → Build Appointment Confirmation / Send Staff New Appointment Email → Respond to Vapi - Booking
manage_appointment	Manage Appointment Webhook → Validate Manage Request → Read Appointments for Lookup → Find Matching Appointment → Appointment Found → Is Cancel Action → Mark Appointment Cancelled / Merge Reschedule Fields → Update Appointment Details → Respond to Vapi - Manage
request_human_callback	Callback Request Webhook → Normalize Callback Data → Log Callback to Sheet → Send Staff Callback Email / Build Callback Response → Respond to Vapi - Callback

9. Reliability & Security Design

- **Explicit confirmation required** — booking and any change to an appointment is rejected unless the caller's confirmation flag was set, so the agent can't act on an ambiguous or half-finished conversation.
- **Two-factor lookup for changes** — cancelling or rescheduling requires both the exact appointment ID and the phone number on file, preventing one caller from altering another patient's booking.
- **Server-side conflict checking** — availability is verified against the sheet at booking time, not left to the AI to infer, so double-bookings can't slip through.
- **Input validation on every tool** — each webhook checks its required fields before doing anything, and throws a clear error (surfaced to the voice agent) when something is missing.
- **Full audit trail** — every booking, cancellation, reschedule and callback is logged to Google Sheets and emailed to staff, so nothing happens invisibly.

10. Setup Checklist

- Connect a Google Sheets credential and point every Sheets node at your own spreadsheet, with an Appointments tab and a Callbacks tab.
- Connect a Gmail credential and set the hospital staff notification address on the four “Send Staff ...” email nodes.
- Activate the workflow, then copy each webhook's Production URL into the matching Vapi tool: `hospital_check_availability`, `hospital_book_appointment`, `hospital_manage_appointment`, `hospital_request_callback`.
- In Vapi, configure the voice agent's tool definitions so it passes `caller_confirmed` only once the caller has verbally agreed, and passes the appointment ID back to the caller after every booking.
- Test all four tools end-to-end with a live or simulated call: check availability, book, reschedule, cancel, and request a callback.

This workflow was designed and built end-to-end in n8n as part of an AI automation portfolio, demonstrating a voice-agent backend with real-time conflict checking, identity-verified appointment changes and staff notifications.